Data Analyst Fresher Interview Cheat Sheet

Target Companies: Deloitte, Cognizant, Capgemini | Core: SQL, Power BI, Python, Business Analytics

1. The 90-Second Winning Self-Introduction

Script to memorize:
*'Good morning/afternoon [Interviewer]. Thank you for this opportunity. I am Srujan, an aspiring Data Analyst with strong technical skills in **SQL, Python, and Power BI**.*
*Recently, I designed and deployed an end-to-end **Customer Retention & Churn Risk Analytics Engine**. In that project, I designed a star-schema warehouse, wrote advanced SQL CTEs for cohort retention and RFM segmentation, built an automated AI sentiment pipeline in Python, and published an interactive executive cockpit in Power BI.*
I am eager to join [Deloitte / Cognizant / Capgemini] because of your high-impact analytics consulting practice, where I can apply my data modeling, problem-solving, and storytelling skills to deliver measurable business value.'

2. Top Core SQL Questions & Exact Answers

Q1: WHERE vs HAVING

• **WHERE:** Filters individual rows *before* any aggregation happens. Cannot contain aggregate functions (like SUM/AVG).
• **HAVING:** Filters grouped records *after* `GROUP BY` has executed.

```
SELECT department, AVG(salary) FROM employees WHERE status = 'Active' GROUP BY department HAVING AVG(salary) > 75000;
```

Q2: INNER JOIN vs LEFT JOIN

• **INNER JOIN:** Returns only matching records from both tables.
• **LEFT JOIN:** Returns ALL records from the left table, plus matched rows from the right table. Non-matching right rows become NULL.
• *Interview Hook:* 'I use LEFT JOIN when identifying inactive records (e.g. `WHERE Orders.order_id IS NULL` to find customers who never bought).'

Q3: RANK() vs DENSE_RANK() vs ROW_NUMBER()

When ranking identical values (e.g., scores 90, 90, 80):
• **ROW_NUMBER():** Assigns strictly unique sequential numbers: **1, 2, 3**.
• **RANK():** Leaves gaps after ties: **1, 1, 3**.
• **DENSE_RANK():** Does NOT leave gaps: **1, 1, 2**.

Q4: Find the 2nd Highest Salary (2 Methods)

Method A (Subquery):
SELECT MAX(salary) FROM employees WHERE salary < (SELECT MAX(salary) FROM employees);
Method B (Window Function):
WITH Ranked AS (SELECT salary, DENSE_RANK() OVER (ORDER BY salary DESC) as r FROM employees) SELECT DISTINCT salary FROM Ranked WHERE r = 2;

Q5: Top 3 Spenders per Customer (Aggregates & Grouping)

```
SELECT customer_id, SUM(amount) AS total_spent FROM Orders GROUP BY customer_id ORDER BY total_spent DESC LIMIT 3;
```

Golden Rule: Any non-aggregated column in `SELECT` must appear in the `GROUP BY` clause.

Q6: UNION vs UNION ALL

- **UNION:** Merges datasets, removes duplicate rows, and sorts the output (slower).
- **UNION ALL:** Merges datasets while keeping all rows including duplicates (faster).

3. Top Power BI & DAX Essentials

Q1: Calculated Column vs Measure (CRITICAL)

Feature	Calculated Column	Measure
Evaluation	Row-by-row during data refresh	Dynamically on-the-fly at query time
Storage & RAM	Stored permanently in RAM / file	Uses CPU memory on the fly (almost 0 RAM)
Filter Context	Static (does not respond to slicers)	Dynamic (changes when user clicks slicers)
Best For	Slicers, chart axes, row categorizations	KPI cards, aggregated numerical totals

Q2: Why Star Schema over Snowflake Schema in Power BI?

• **Star Schema:** Fact table in the center directly connected to single-level dimension tables ($1:*$). Delivers faster DAX performance, uses simpler relationship paths, and optimizes Power BI's VertiPaq engine.
• **Snowflake Schema:** Dimension tables are normalized into sub-dimensions, requiring multiple table hops and slower joins.

Q3: What does the CALCULATE() function do in DAX?

• `CALCULATE()` evaluates an expression while modifying, overriding, or adding new filter context. It is the most powerful function in DAX.

4. Python for Analytics Basics

• **Handling Missing Data:** `df.isna().sum()` (detect), `df.dropna()` (drop), `df.fillna(value)` (impute with mean/median).
• **Merging Data:** `pd.merge(df1, df2, on='customer_id', how='left')`.
• **Aggregation:** `df.groupby('channel')['amount'].sum().reset_index()`.

5. Behavioral & Closing Strategy (The STAR Method)

When answering scenario questions (e.g. 'Tell me about a challenge with data'):
• **S (Situation):** Set the background in 1 sentence.
• **T (Task):** The goal you needed to achieve.
• **A (Action):** The specific technical action you took (SQL/Power BI/Python).
• **R (Result):** The measurable outcome (% improvement, time saved).

Smart Questions to Ask the Interviewer at the End:

1. *'What does the typical analytics tech stack and data maturity look like on the engagements I would support?'*
2. *'What qualities differentiate a good fresher analyst from a great one on your team?'*